Problem 6

Problem 6

Problem 1 *Find all points at which the given function equals its average value on the given interval.*

(a) $f(x) = e^x \qquad [0, 4]$

Explanation. First, we need to find f_{avg}:

$$\begin{aligned} f_{\text{avg}} &= \frac{1}{4-0}\int_0^4 e^x\,dx \\ &= \frac{1}{4}\Big[e^x\Big]_0^4 \\ &= \frac{1}{4}\left(e^4 - 1\right) \end{aligned}$$

So we are looking for all values $c \in [0, 4]$ such that:

$$\begin{aligned} f(c) &= \frac{1}{4}(e^4 - 1) \\ \implies \quad e^c &= \frac{1}{4}(e^4 - 1) \\ \implies \quad c &= \ln\left(\frac{1}{4}(e^4 - 1)\right) \end{aligned}$$

Therefore, our answer is $\ln\left(\frac{1}{4}(e^4 - 1)\right)$.

(b) $g(x) = \frac{\pi}{4}\sin(x) \qquad [0, \pi]$

Explanation. First, we need to find g_{avg}:

$$\begin{aligned} g_{\text{avg}} &= \frac{1}{\pi-0}\int_0^\pi \frac{\pi}{4}\sin(x)\,dx \\ &= \frac{1}{4}\Big[-\cos(x)\Big]_0^\pi \\ &= \frac{1}{4}\left(-(-1-1)\right) \\ &= \frac{1}{2} \end{aligned}$$

So we are looking for all values $c \in [0, \pi]$ such that:

$$\begin{aligned} & g(c) = \frac{1}{2} \\ \implies \quad & \frac{\pi}{4}\sin(c) = \frac{1}{2} \\ \implies \quad & \sin(c) = \frac{2}{\pi} \\ \implies \quad & c = \arcsin\left(\frac{2}{\pi}\right), \pi - \arcsin\left(\frac{2}{\pi}\right) \end{aligned}$$

Therefore, our two answers are $\arcsin\left(\frac{2}{\pi}\right), \pi - \arcsin\left(\frac{2}{\pi}\right)$.